

Proton Treatment of Uveal Melanomas

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Proton beam radiation therapy (PBRT) of uveal melanomas was introduced in the treatment of intraocular tumours nearly 30 years ago by the group at the Harvard Cyclotron [8]. It is widely accepted that with respect to basic physical data PBRT offers distinct advantages including homogenous dose distribution, minimal beam scatter, the opportunity of fractionation and a sharp dose fall below the target volume due to the physical properties of the Bragg-peak.

PBRT, however, is extremely expensive as a cyclotron with a minimal energy of at least 55 MeV is needed for the treatment of eye tumours [7]. This means it is not a low cost treatment installation.

However, PBRT has a few disadvantages not only from the economic but also from the physical point of view. Radiation damage to the ocular surface and ocular adnexa is much more pronounced than in any of the brachytherapy techniques. As in the pioneering days of PBRT, treatment planning is still based on the use of tantal clips used as orientation markers detectable with x-rays for three-dimensional reconstruction of the globe and identification of radiosensitive structures like the macula, the optic nerve and the intraocular mass. After 30 years of PBRT, there is virtually no randomized study comparing proton and radioactive plaques in the management of uveal melanomas; to date a single study has been published on the use of helium ions versus iodine-125 plaques [2]. The only randomized study available for PBRT of uveal melanomas looked exclusively for different radiation doses in PBRT; it did not look for the comparison of different radiation techniques. Consequently all data currently available for the efficacy of PBRT of uveal melanoma have a low evidence level since nearly all studies published are single-centre, retrospective, non-randomized and not even matched studies. As a result, published data on PBRT of uveal melanoma are controversial and cannot be compared to other radiation techniques.

Favourable and very unfavourable findings have been published on the results of PBRT of uveal melanomas. Some centres have published eye retention rates as high as 100% in subgroups of choroidal melanomas [4], which would be unique in the treatment of solid tumours. Looking at the long-term results

of PBRT of posterior pole choroidal melanomas and of large uveal melanomas, must be substantial doubt as to whether the theoretical advantages of PBRT may be implemented in real life and PBRT has an advantage over sophisticated 106-ruthenium plaque radiotherapy [1, 6, 11]. There is fundamental dissent in the indications for PBRT of uveal melanoma as PBRT is practically the only treatment modality for uveal melanoma in some centres, while other centres treating large numbers of patients have never used PBRT [10]. Data on eye retention rates, local recurrences, radiation-related complications and functional results in terms of preservation of vision are limited to retrospective single-centre series. No data on regression patterns or patterns of tumour recurrences in comparison to other eye-salvaging treatment modalities have been evaluated to date.

In view of this dilemma, a discussion on the incidence and classification of recurrences and the pattern of tumour regression after PBRT as published in the current issue of this journal was inevitable and overdue. Tribute should be paid to M. Fitzek [5] for initiating this discussion and pointing out that the classification of tumour control and recurrence after PBRT of uveal melanomas may be inconclusive within the same series. In response, S. Höcht et al. [9] have likewise made clear that their criteria for classifying uveal melanoma regression patterns have changed as they have learnt that an increase in maximum tumour thickness does not always reflect tumour growth. This discussion emphasizes the need for a well-designed, prospective, definitely randomized and preferably multi-centred study on the efficacy of PBRT of uveal melanomas in comparison to other radiation modalities thus establishing a classification of tumour growth and regression pattern. The Collaborative Ocular Melanoma Study (COMS) has proved that despite all problems related to long recurrence intervals and other epidemiological peculiarities of uveal melanomas studies of this nature may be carried out. Such a study would certainly add substantially to our clinical understanding of uveal melanomas [3]. In view of spiralling health cost in all western countries and as PBRT is an extremely costly treatment, this study is essential from a scientific point of view and – perhaps even more urgently – in terms of public health financing.

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